

Mini Project: Real Estate Insights Dashboard

Project Background

You have been contracted by the Municipal Planning and Zoning Board to analyze and visualize real estate sales data for the region. Over the past decade, property transactions have fluctuated significantly across towns and property types.

The board needs data-driven insights to inform taxation policy, zoning revisions, and housing development strategy. Your task is to design an interactive Tableau dashboard that helps stakeholders explore these patterns clearly and intuitively.

Objective

Build a Tableau-based analytical dashboard that highlights real estate trends, identifies top-performing towns, and provides comparative insights into property values and sales ratios.

By the end of this project, you will demonstrate your ability to:

- Clean and transform real-world data for visualization.
- Design interactive dashboards with calculated fields and parameters.
- Translate complex data into actionable insights for decision-makers.

Dataset Overview

You are provided with a real estate transactions dataset covering **2011–2022**. Each record represents a single property sale.

Key Fields:

- Serial Number — Unique transaction ID
- List Year — Year property was listed
- Date Recorded — Official sale recording date
- Town — Municipality where the sale occurred
- Address — Property address
- Assessed Value — Government-assessed property value
- Sale Amount — Final sale price
- Sales Ratio — Ratio of assessed to sale value

- Property Type — (e.g., Residential, Commercial, Industrial)
- Residential Type — (e.g., Single-Family, Condo, Apartment)
- Non Use Code, Assessor Remarks, OPM Remarks — Administrative notes
- State — State abbreviation

Project Phases

Phase 1 – Data Preparation & Exploration

- Connect the Excel dataset (Real_Estate_Sales_2011-2022_GL_1.xlsx) in Tableau.
- Inspect field names, data types, and null values.
- Create derived fields such as:
 - Sale Year (from Date Recorded)
 - Sale Category (High / Medium / Low — based on thresholds)
- Summarize data using simple visual explorations (e.g., distribution of sale amounts).

Phase 2 – Visual Exploration

Create the following foundational charts to explore the dataset:

Visualization	Description
Bar Chart	Total Sale Amount by Town
Line Chart	Year-over-Year or Month-on-Month Sales
Pie Chart	Distribution of Property Types
Highlight Table	Average Sales Ratio by Town and Residential Type
Treemap	Volume of Sales by Residential Type

Each chart should have meaningful titles, labels, and color schemes for interpretability.

Phase 3 – Mapping & Time Analysis

- Build a map visualization showing sales distribution by *State* or *Town*.
- Add a time-series chart displaying monthly or yearly sales trends.
- Use filters and tooltips to allow regional comparisons over time.

Phase 4 – Calculated Fields & Parameters

- Create a calculated field to categorize Sale Amount (e.g., *High*, *Medium*, *Low*).
- Add a parameter control that lets the user toggle between key metrics:
 - Sale Amount
 - Assessed Value

This allows users to explore the dataset through multiple financial lenses.

Phase 5 – Dashboard Design

Combine your visualizations into an interactive dashboard with:

- Filters for Town, Year, and Property Type
- A highlight action (e.g., hover to emphasize a town or property category)
- A filter action (e.g., click on a chart element to filter others)
- A URL action that dynamically links to a web page using the Town name (for example, a Wikipedia page or real estate portal for that town)

Design the dashboard for clarity and storytelling, not clutter.

Phase 6 – Storytelling & Publishing

- Create a Tableau Story with 3–4 slides that narrate your insights.
- Example storyline:

1. *Overall Trends By Year*

2. *Town-wise Performance & Sales Ratio*
3. *Property Type Composition*
4. *Key Insights & Recommendations*

Deliverables

1. A packaged Tableau workbook (.twbx) with all data and dashboards.
2. A short summary or storyboard explaining 3–5 key findings from your analysis.